

Premises: 1) $(\neg p \vee \neg q) \rightarrow (r \wedge s)$ 2) $r \rightarrow \neg t$ 3) $\neg t$

Conclusion: p

Steps	Reason
1) $r \rightarrow \neg t$	Premise
2) $\neg t$	Premise
3) $\neg r$	Modus tollens on 2, 1
4) $\neg p \vee \neg s$	Addition on 3
5) $\neg(r \wedge s)$	De Morgan's law on 4 (1st)
6) $(\neg p \vee \neg q) \rightarrow (r \wedge s)$	Premise
7) $\neg(\neg p \vee \neg q)$	Modus tollens on 5, 6
8) $\neg(\neg(p \wedge q))$	De Morgan's law on 7 (1st)
9) $(p \wedge q)$	Double-negation law on 8
10) p	Simplification on 9

7) What rules of inference are used in this famous argument? ~~Normal~~
 "All men are mortal. Socrates is a man. Therefore, Socrates is mortal."

Soln: $Ma(x): x$ is a Man $Mo(x): x$ is Mortal

Common Domain: Set of all people

Premise: 1) $\forall x (Ma(x) \Rightarrow Mo(x))$ 2) $Ma(\text{Socrates})$

Conclusion: $Mo(\text{Socrates})$

Steps	Reason
1. $\forall x (Ma(x) \Rightarrow Mo(x))$	Premise
2. $Ma(\text{Socrates})$	Premise
3. $Ma(\text{Socrates}) \Rightarrow Mo(\text{Socrates})$	Universal instantiation on 1
4. $Mo(\text{Socrates})$	Modus ponens on 2, 3